



The THB Scenario Builder

Methodology: what the tool calculates, where every number comes from, and what it does not represent

6

assumptions

7

CMIP7 scenarios

6

Learn More pages

75

years, 2025 to 2100

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This is work in progress — caveat lector.

The tool this documents runs at scenarios.thehonestbroker.org. Every figure it reports is recomputed from the sources named in Part 3 rather than carried in prose, and every rate is rebuilt by a script that reads the primary series.

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Global Carbon Budget
ScenarioMIP CMIP7 marker runs
UN World Population Prospects 2024
IPCC 2006 Guidelines, default carbon content
Maddison Project Database 2023
IIASA SSP database v3.2 (June 2025 release)
FaIR v2.2

The files

base.json
observed.json
learn_population.json
learn_energy_intensity.json
learn_fuel_mix.json
learn_income.json
learn_methane.json
learn_land_use.json
analogues.json
markers.json, population.json, emulator.json, config.json, presets.json, notes.json

Vintages

Part 1

The site

An overview of what the THB Scenario Builder does, page by page, and how to read what it puts on the screen. The two parts that follow this one give the detail: **Methods** covers every calculation and every preset, **Data** covers where each number came from, when, and in what units.

What the tool does

The builder turns six assumptions into one emissions path from 2025 to 2100, draws that path against the seven CMIP7 marker scenarios, and reports what the path adds up to.

Nothing about the exercise depends on agreeing with any of the six numbers. The point is the arithmetic that connects them: a reader who thinks the world will decarbonise faster than the record can set that rate and watch what it does to 2100, on the same axis and against the same seven published scenarios as everybody else.

The six assumptions

Four of them multiply together in the Kaya identity, and two sit outside it:

Slider	Sets	Range	Today
People in 2100	A level	6 to 14 billion	8.14 billion
Income growth per person	A rate that compounds	−0.5 to +3.5 %/yr	\$21,393 in 2024
Energy needed per dollar	A rate that compounds	−4 to +0.5 %/yr	3.40 MJ per dollar
Carbon per unit of energy	A rate that compounds	−4 to +0.5 %/yr	65.18 kgCO2 per GJ
Land use CO2 in 2100	A level	−10 to +5 GtCO2/yr	3.83 GtCO2/yr
Methane in 2100	A level	80 to 600 Mt/yr	380 Mt/yr

The first four decide fossil and industrial CO2, cement included. Land use CO2 adds on top as a separate term, running in a straight line from today to whatever the reader sets. Methane changes nothing about the CO2 path and enters only through the warming figure.

The base year is 2025, carrying the observed 2024 world, because 2025 has not closed in every source.

What the chart shows

One heavy line for the reader's scenario, seven lighter lines for the CMIP7 markers, and annual CO2 including land use on the vertical axis.

That axis moves. Every render picks an axis that covers the reader's path and all seven markers together, on round steps, with zero always on it. A scenario far above the published range stays fully drawn instead of clipping, and the seven markers never leave the axis the reader's own line sits on.

Whatever the reader types into **Name your scenario** names the line on the chart, the column in the table, and every file the page hands back.

The four figures under the chart

- **Cumulative CO2, 2025 to 2100.** Every year of the path, summed, in gigatonnes.
- **Warming in 2100 above 1850-1900.** A curve fitted to FaIR v2.2 runs of the seven markers, reading cumulative CO2 and 2100 methane and nothing else. Indicative, not a model result, and never shown to more than two decimals.
- **Added warming from now.** The same figure measured against the 2015-2024 average of 1.24 °C rather than against 1850-1900.
- **Your 2100 world looks like.** The largest economy whose 2024 CO2 per dollar of GDP sits within 5% of the reader's 2100 world, drawn from 66 economies large enough to measure well. Largest, rather than nearest, so the comparison lands on an economy the reader recognises instead of whichever small one happens to sit on the number. Where nothing sits within 5% the tile says so instead of naming an economy, and names the one at the edge of the table instead: a world at 0.011 kg CO2 per dollar resembles nothing on earth today. It compares that one ratio and says nothing else about the country.

Load one of the four CMIP7 presets and the first three figures report what that scenario published, rather than the reconstruction, until a slider moves.

The eight preset buttons

Four presets come from the record and from the two technology bounds; four load the Kaya factors a CMIP7 marker reports. Those four are the markers of the seven

that publish a carbon-intensity rate.

Preset	Cumulative CO2	Warming in 2100
Kaya at observed rates	4,178 Gt	3.21 °C
Trend continues	3,409 Gt	2.94 °C
Slowest technical progress	5,047 Gt	3.47 °C
Ausubel methane economy	1,585 Gt	2.22 °C

Every rate behind those four comes out of `scripts/build_data.py`, recomputed from the primary series rather than typed in, and the window that produced each one travels with it.

The four CMIP7 presets reproduce where a marker ends up far better than how it gets there, and three of them cannot reach the scenario they name at all. The interface reports the gap in place, computed on the spot, whenever one of them loads. **Methods** gives the three reasons — the shape of the path, the sign of the emissions, and the timing of the land-use sink — and the size of each miss. MEDIUM-to-LOW is the one that lands below its marker rather than above.

The Learn More pages

Six pages, one per assumption. Each teaches the quantity, shows what the world has done, and hands back a value the reader built rather than guessed:

- **Population** adds up seven regional 2100 populations, each running between the UN's low and high variants.
- **Income per person** works in any of three directions: set a rate, set a 2100 level, or build the world average out of the three World Bank income groups.
- **Energy per dollar** takes a compound rate from any two years of the record, or a multiple of the observed rate, and says where the answer falls among the 35 twenty-five-year windows the world has actually run.
- **CO2 per unit of energy** builds a 2100 fuel mix out of seven shares, burns it against the IPCC emission factors, adds the industrial CO2 that no fuel switch touches, and takes the rate that lands there.
- **Land use** nets four flows — deforestation, other transitions and peat, existing regrowth, restored area — plus engineered removal.
- **Methane** sets one control per source, scaled so today's five sources reproduce the 380 Mt the slider starts from.

A page shows the arithmetic at every step, not just the answer. **Use in my scenario** carries the value back to the builder, which names it, scrolls to that slider and lights

it; **Back to my scenario** returns the six numbers untouched.

What travels with a scenario

Six numbers and a name, encoded in the address. Every link off every page carries them, so a reader who steps out to a Learn More page, the library or the bibliography comes back to the scenario they left. **Copy link to this scenario** hands over that same address.

The builder downloads a PNG and a PDF of the whole scenario sheet. Every figure on every page downloads two ways: as a PNG carrying the figure's title, the drawing and the numbers under it, and as a spreadsheet of those numbers. Where a figure rests on more rows than an image can hold, the PNG prints an evenly spaced sample and says so; the spreadsheet always carries every row.

What every figure carries

The Honest Broker mark, the source of the numbers, and the analysis credit, in that order, under every chart and every table on the site. The downloaded PNG draws the same three underneath the figure and the spreadsheet writes the last two under the table, because a figure that travels has to answer for itself.

The library and the bibliography

The **library** collects the posts at The Honest Broker that make the argument this tool grew out of, grouped by what each one argues. The **bibliography** lists every work the site draws on, each with a link and a plain-language note on what it supplied, and it generates the Learn More section of that list from the pages themselves, so a source cited on a page cannot go missing from the list.

What the tool does not do

Stated in full in **Methods**, and worth having up front:

- Nothing engineered removal does can come out of the four Kaya factors, which stay positive whenever there are people, income and energy. The land use slider carries the only sink, and it stands in for land and engineered removal together.
- No nitrous oxide, no fluorinated gases, no aerosols as a term of their own.
- No carbon cycle feedbacks that depend on the path rather than the total.

- One warming number, not a range. The FaIR ensemble behind the fit spans well over a degree at any given cumulative total.
- Only 2100, not the path of warming to it or anything after it.
- No check that six assumptions belong together. A handful of combinations no marker contains get named as an observation, not blocked.

How to check any of it

Every number the interface shows comes from a JSON file under `src/data/`, and nothing gets typed into the interface code. **Data** names the script that writes each file, the source it reads, and the vintage of that source. The scripts that can rebuild from a primary source say so; the few series carried over from the prototype say that instead, and name what would verify them.

Part 2

Methods

What the tool calculates, how the four presets were derived, and what it does not represent. Part 3 covers where the numbers come from.

The emissions calculation

Emissions follow the Kaya identity. Four factors multiply together, and land use CO2 is added on top as a separate term:

$$\text{CO2} = \text{population} \times (\text{GDP} \div \text{population}) \times (\text{energy} \div \text{GDP}) \times (\text{CO2} \div \text{energy}) \\ + \text{land use CO2}$$

The reader sets six numbers: the 2100 population, three annual rates that compound from the base year, and 2100 values for land use CO2 and methane.

The base year is 2025, holding the observed 2024 state, since 2025 is not complete in every source. The path runs one year at a time to 2100 and the cumulative total sums every year.

What the four factors cover

They cover **all fossil and industrial CO2, cement included**. This matters, because the whole chart is the reader's line drawn against the seven CMIP7 marker lines, and those count all of it.

The prototype set base CO2 per unit of energy from the Energy Institute's energy CO2 plus flaring, at 60.5 kg per GJ. That leaves out cement and other industrial process emissions, roughly 1.9 GtCO2 a year, and it started every path at 38.70 GtCO2 in 2025 while the markers started between 42.4 and 43.8. On 2026-09-04 the base was recalibrated to 65.181 kg per GJ, taken from Global Carbon Budget fossil and industry CO2 over Energy Institute total energy supply. Paths now start at 42.47, inside the marker range.

This changed results. Kaya at observed rates moved from 3,707 GtCO2 to 4,084, which puts it above CMIP7 HIGH's 3,777 rather than below it. `presets.json` keeps the figures stated before the recalibration under `brief_stated`, and a test pins the size of the shift.

Population

Population interpolates between the real IIASA SSP1, SSP2 and SSP3 world trajectories on their 2100 endpoints, so the curve keeps a demographic shape: SSP1 peaks at 9.29 billion around 2060 and declines, SSP3 rises throughout. A straight line or a smoothstep between two endpoints could not do that.

Outside the SSP1 to SSP3 span, which is 8.09 to 12.98 billion in 2100, the nearest curve is scaled by the ratio of the target to its own endpoint. That keeps the shape and moves the level, and it also moves the 2025 end of the curve: at the extremes of the slider the path starts from 6.04 or 8.79 billion rather than 8.15. Both ends of the 6 to 14 billion slider reach this. A test pins the behaviour so it cannot change unnoticed.

Land use

Land use CO2 runs in a straight line from 3.83 GtCO2 today to whatever the reader sets for 2100.

A line is a placeholder. The marker paths carry land use folded into their total CO2 and publish no separate land use series here, so there is no shape to borrow from them. If those series become available, the shape should follow them.

Warming

Warming in 2100 above 1850 to 1900:

$$T = 1.7274 + 1.3100 \times \ln(1 + \max(0, C) \div 1989.74) + 0.0012 \times (CH4 - 380)$$

c is cumulative CO2 from 2025 to 2100 in gigatonnes and CH4 is methane in 2100 in megatonnes a year. Methane adds about 0.12 °C per 100 Mt a year.

This is a curve fitted to FaIR v2.2 runs of the seven CMIP7 markers, using the fair-calibrate v1.4.1 constrained ensemble. **It is indicative, not a model result**, and the interface says so. Warming is never shown to more than two decimal places.

How well it reproduces FaIR

Scenario	Cumulati ve CO2 (Gt)	Methane (Mt/yr)	Emulator	FaIR	Error
VERY LOW	268	94.0	1.55	1.61	−0.06
LOW-to-NEGATIVE	373	216.0	1.76	1.68	+0.08
LOW	605	150.8	1.80	1.83	−0.03
MEDIUM-to-LOW	1,710	203.4	2.33	2.20	+0.13
HIGH-to-LOW	2,524	132.8	2.50	2.76	−0.26

Scenario	Cumulative CO2 (Gt)	Methane (Mt/yr)	Emulator	FaIR	Error
MEDIUM	2,770	436.2	2.94	2.84	+0.10
HIGH	3,777	533.1	3.31	3.26	+0.05

Root mean square error 0.12 °C. Six of the seven fall within 0.15 °C. HIGH-to-LOW does not.

What the emulator cannot see

It reads cumulative CO2 and methane, and nothing else. Two paths reaching the same total give the same answer however differently they got there. HIGH-to-LOW is where that bites: it climbs to 51.6 GtCO2 a year in 2065, falls steeply after that, and carries the second-lowest methane of the seven. A fit with no term for the shape of the path has no way to tell it apart from a scenario that arrives at 2,524 Gt smoothly, and it lands 0.26 °C low.

Refitting for one structural outlier would move numbers already in print for the sake of a case the functional form cannot represent. The miss is documented and pinned by a test instead.

The four presets

Each loads all six inputs. The two bounds follow Ausubel's 1995 argument that technological trajectories move at rates steady enough to bound the future, and that a scenario halting them describes technical regression rather than business as usual. **Neither bound assumes poverty. Neither assumes a technology stops working.**

Preset	Population	Income	Energy/\$	CO2/energy	Land use	Methane	Cumulative	Warming
Kaya at observed rates	10.2	+1.91	−1.43	−0.15	2.0	380	4,178	3.21
Trend continues	10.2	+1.91	−1.62	−0.48	1.0	300	3,409	2.94
Slowest technical progress	11.4	+2.23	−1.36	−0.17	2.0	450	5,047	3.47
Ausubel methane economy	9.0	+1.91	−1.57	−2.79	−1.0	150	1,585	2.22

Kaya at observed rates. The 1990 to 2024 record projected forward unchanged. Income +1.91, energy per dollar −1.43 and CO2 per unit of energy −0.15 all come straight out of the Energy Institute, World Bank and Global Carbon Budget series. The carbon-intensity figure was −0.21 until 2026-09-05, on a narrower basis; "The

corrected carbon-intensity rate" below gives the change and what moved with it.

Trend continues. The 2015 to 2024 rates, faster than the long record on both technology terms: −1.62 and −0.48. Income stays at the long-record 1.91 rather than the recent decade's 2.02.

Slowest technical progress. Every technological trajectory held at the slowest sustained rate on record. Energy per dollar at −1.36 %/yr, the weakest 25-year window since 1990, which is 1990 to 2015. The fuel mix at −0.17 %/yr, the weakest 30-year window since 1965, which is 1992 to 2022. Income at +2.23, the fastest observed 25-year rate, 1994 to 2019, and population at 11.4 billion, the top of the UN 2024 95% range, so emissions run as high as slow technology permits.

Ausubel methane economy. Jesse Ausubel's 1988 published trajectory, which squeezes carbon out of primary energy to 0.06 tonnes of carbon per kilowatt-year by 2100, implying the fuel mix improving 2.79 %/yr. Energy per dollar at −1.57, its fastest observed 25-year rate, 1996 to 2021, and income still growing at the historical pace.

Every rate above is recomputed by scripts/build_data.py from the primary sources rather than typed in, and the window that produced each one is recorded alongside it.

The four CMIP7 presets

Four further buttons load the Kaya factors that a marker itself reports. These are the four markers of the seven that publish a carbon-intensity rate; the other three do not, because their 2100 fossil CO2 runs negative. HIGH lands close to the scenario it names. The other three cannot:

Preset	Marker total	Preset reproduces	Gap
CMIP7 HIGH	3,777 Gt	3,838 Gt	+61
CMIP7 MEDIUM	2,770 Gt	3,094 Gt	+324
CMIP7 MEDIUM-to-LOW	1,710 Gt	1,230 Gt	−480
CMIP7 VERY LOW	268 Gt	1,298 Gt	+1,030

MEDIUM-to-LOW is the one that lands below the marker it names rather than above. The reason is in the next section. The interface says so in place when one of these is loaded rather than leaving the reader to notice.

The Learn More pages

Each of the six assumptions gets a page that teaches the quantity, shows the record, and hands the reader a value built from assumptions they control. The pages share one scaffold (`src/ui/learn/page.ts`), one builder (`src/ui/learn/builder.ts`) and one plot component (`src/ui/plot.ts`), and each page module supplies only its own words, chart and arithmetic.

What every figure carries

The mark, the source of its numbers and the analysis credit, in that order, under every chart and every table on every page. The downloaded PNG draws the same three in a band beneath the figure, and the downloaded workbook writes the source and the credit under the table, because a figure that travels has to answer for itself.

The hand-off

A link on the top page carries the reader's scenario to a page as `/learn/<slug>/?s=<six numbers>&n=<name>`, the same encoding share links use. The page holds that string untouched. Two exits lead back:

- **Back to my scenario** returns `/#s=<the identical string>`.
- **Use in my scenario** returns `?applied=<field>#s=<state with one field replaced>`. The top page names the value, scrolls that slider into view, lights it for 1.5 seconds, and then rewrites the address bar to a clean `/#s=...` so a copied link carries no stale message.

A builder value passes through `roundToStep` and then `clampWithFlag` (`src/model/config.ts`), and the page prints the rounded value, and the clamp when one bites, before the reader commits. No page duplicates any of that arithmetic.

The population builder

Seven regional 2100 populations, added up:

```
world 2100 = Σ region 2100
```

Each control opens at the UN's medium variant and runs from the UN's low variant to its high variant. The UN builds those two by subtracting and adding half a child per woman at every date, and they add up across regions, so the sum reproduces the UN's own world figures exactly: 6.987, 10.180 and 14.395 billion.

`scripts/build_wpp.py` asserts that at build time.

Three consequences worth stating, all of them visible on the page:

- 1 The builder cannot reach the slider's floor of 6 billion. Every region at the UN's low variant still gives 6.99 billion.
- 2 It can exceed the slider's ceiling of 14 billion, which reports 14.395 and clamps to 14.

- 3 The 95% prediction intervals do not add up this way, and the page says so rather than summing them: the regional lower bounds give 8.234 billion against the UN's world lower bound of 9.047.

The reader's curve on that page comes from `populationAt` in `src/model/population.js`, the same function the scenario itself uses, so the shape on the Learn More page and the shape behind the top page agree.

The energy per dollar builder

Two modes, both producing a rate in %/yr.

A window from the record. The reader picks two years and the builder takes the compound rate between the observed intensities of those years:

```
rate = ((intensity[end] / intensity[start]) ^ (1 / (end - start)) - 1) x 100
```

The second year is held at least ten years after the first, because a shorter window measures a business cycle rather than a trend.

A multiple of the observed rate. The rate observed from 1990 to 2024, -1.4277 %/yr, times a multiple the reader sets between 0 and 3.

Both modes report where the answer falls among the 35 windows of 25 years in the record, as a count and as a percentile, and what today's 3.40 MJ per dollar becomes by 2100 at that rate. The percentile counts windows that improved more slowly than the chosen rate, so a faster rate scores higher.

The series behind all of it splices two sources at 1990; Part 3 gives the method and the size of the difference between them.

The CO2 per unit of energy builder

Seven fuel shares for 2100 plus one control for industrial CO2, in four steps:

- 1 **Normalise.** The seven shares are scaled to sum to 100, so only their proportions matter.
- 2 **Burn.** Each share meets its emission factor, and the sum is scaled by 0.9201, the calibration that makes the 2024 mix reproduce the 2024 observed intensity. Part 3 explains where that factor comes from.
- 3 **Add what does not burn.** 5.237 kgCO2 per GJ of cement, flaring and other industrial CO2, times whatever the last control is set to.
- 4 **Take the rate.** From 65.181 kgCO2 per GJ today to that number in 2100:

```
rate = ((intensity[2100] / 65.181) ^ (1 / 75) - 1) x 100
```

The page shows every intermediate number, because the interesting part of this builder is the arithmetic rather than the answer. A mix with no fossil fuel in it still

leaves step 3, so the fastest rate reachable with industrial CO2 unchanged is $-3.31\%/yr$.

The income per person builder

Three modes.

Set a rate. The level in 2100 is $21,393.2 \times (1 + \text{rate}/100)^{75}$.

Set a 2100 level. The rate is the inverse: $((\text{level} / 21,393.2)^{(1/75)} - 1) \times 100$.

By income group. Each group's average compounds at its own rate from its own 2024 level, and the world average is the population-weighted sum, scaled by 0.9967 so the three groups reproduce today's world figure. Population shares stay at today's values, which the page states, along with the direction of the bias: the UN projects the low-income share rising, so holding shares still overstates the world average a little.

The methane builder

One control per source, added up. Each source's 2024 level is scaled by 1.086, the ratio between the tool's 380 Mt base year and EDGAR's 350 Mt inventory total, so today's five sources reproduce the number the slider starts from. Part 3 explains why the two differ and why both sit inside the published range.

The page also reports what the answer is worth in the emulator, at 0.12°C per 100 Mt against a reference of 380 Mt, and says plainly what that coefficient cannot represent: it comes from a straight-line fit to seven FaIR runs and ignores when the methane is emitted.

The land use builder

Four flows netted into one 2100 figure:

```
net = (deforestation + other transitions and peat) x scale
      - existing regrowth x scale
      - restored area x sequestration rate
      - engineered removal
```

with today's flows from the Global Carbon Budget: 6.23 GtCO₂ of gross deforestation, 2.20 of other transitions and peat, and 4.40 of regrowth. The restoration term is a straight area-times-rate product, in millions of hectares times tonnes of CO₂ a hectare a year, divided by a thousand to reach GtCO₂.

Engineered removal sits on this control because a product of four positive factors cannot go below zero. Every marker that reaches net negative CO₂ does it through terms that land on this line.

What a CMIP7 preset does and does not reproduce

Loading a CMIP7 preset sets the six sliders to the Kaya factors that marker reports. Compounding those factors at a constant rate reproduces where the marker ends up far better than how it gets there, and the interface now reports both, computed rather than asserted (markerFidelity in src/model/flags.ts).

Preset	2100 CO2, this tool	Marker	Cumulative, this tool	Marker	2050, this tool	Marker
CMIP7 HIGH	55.9	55.0	3,838	3,777	48.8	47.1
CMIP7 MEDIUM	34.0	34.4	3,094	2,770	43.3	36.1
CMIP7 MEDIUM-to-LOW	−2.9	−9.2	1,230	1,710	22.4	35.1
CMIP7 VERY LOW	−0.1	−5.8	1,298	268	22.5	−1.2

Three separate causes, and the interface names whichever applies:

Shape. A constant rate spreads one improvement evenly across 75 years, while the markers bend. MEDIUM lands within 1% of its own 2100 emissions and still accumulates 12% more over the century, because MEDIUM cuts hardest in the 2030s and 2040s. Reproducing that would need the markers' own factor trajectories decade by decade, which the marker files here do not carry; see Part 3.

Sign. VERY LOW removes more CO2 than it emits from around mid-century. Four factors multiplied together stay positive, so the fossil term cannot turn negative at all, and only the land use slider can pull a path below zero. The tool reproduces the descent as far as the point where VERY LOW's own emissions cross zero and no further.

Land use timing. MEDIUM-to-LOW ends the century with a land-use sink of −8.8 GtCO2 a year, by far the largest of the seven. The land use slider draws a straight line from today's source to that sink, so the tool banks the sink from 2025 while the marker builds it late, and the same shape error that lifts MEDIUM above its marker pushes MEDIUM-to-LOW 480 Gt below its own. It is the one preset that undershoots. tests/presets.test.ts pins the sign as well as the size, so a change that flipped it could not pass quietly.

The chart brings the named marker's own published path forward whenever one of those presets is loaded, so the divergence sits in front of the reader rather than in a footnote.

The corrected carbon-intensity rate

Until 2026-09-05 the mark under the CO2-per-energy slider read -0.21% /yr, which measures CO2 from energy over total energy supply. The slider itself moves a wider quantity: the fossil and industrial CO2 the CMIP7 markers count, cement included. On that basis the world improved -0.15% /yr from 1990 to 2024, and the mark, the observed rate the notes compare against, and the "Kaya at observed rates" preset now all use it.

What moved with it:

	Before	After
Mark under the slider	-0.21% /yr	-0.15% /yr
"Kaya at observed rates", cumulative	4,083.9 GtCO2	4,177.7 GtCO2
"Kaya at observed rates", warming	3.19 °C	3.21 °C

config.json keeps the old figure under supersededRates and the preset keeps its old totals under expected.superseded, so anything published against the earlier numbers can be traced. The two technology bounds still use window rates measured on combustion alone, which observed.json derives from the Energy Institute series; moving those would change the bounds themselves, and that has not been done.

How far the marker factors can be checked

The seven markers' Kaya factors arrived with the prototype and have no primary source on this machine; verifying them needs the IIASA ScenarioMIP database, which requires an account. Two checks are possible without it, and both were run.

Populations reconcile exactly. HIGH takes SSP3's 12.98 billion, MEDIUM and three others take SSP2's 9.89, HIGH-to-LOW and VERY LOW take SSP1's 8.09 to 8.12. Those match population.json to the last digit.

The rates reconcile with each marker's own 2100 total, where a rate can exist. Compounding population, income and energy per dollar from the base year, then solving for the carbon intensity rate that lands on the marker's own 2100 fossil CO2:

Marker	Published rate	Rate implied by its own 2100 total
HIGH	-0.08% /yr	-0.10% /yr
MEDIUM	-0.87% /yr	-0.85% /yr
HIGH-to-LOW	-2.78% /yr	-3.50% /yr
MEDIUM-to-LOW	-3.06% /yr	no rate reaches it
LOW, LOW-to-NEGATIVE, VERY LOW	none published	no rate reaches it

HIGH and MEDIUM agree within 0.02 percentage points, which is as much corroboration as an internal check can give. HIGH-to-LOW is 0.72 points out.

The last four rows are the interesting result. Those four markers reach 2100 with fossil and industrial CO₂ **below zero** once their land-use term is separated: −0.4, −3.7, −23.2 and −1.1 GtCO₂ a year. A rate applied to a positive quantity cannot reach a negative one, which explains why three of them publish no carbon-intensity rate at all. It is the same structural limit the CMIP7 preset note reports on the front page.

What the tool does not represent

Engineered carbon removal. The four factors multiply to a positive number whenever there are people, income and energy. Nothing in them can go below zero, so a scenario that removes more carbon than it emits cannot be built here. Five of the seven markers reach net negative CO₂, VERY LOW from 2050 and HIGH-to-LOW only in 2100, and the low ones rely on it. The land use slider reaches −10 GtCO₂ a year, which is the only sink the tool has, and it stands in for land and engineered removal together.

Gases other than CO₂ and methane. No nitrous oxide, no fluorinated gases.

Aerosols. Sulphate and other aerosols cool, and scenarios that burn less coal lose that cooling as well as the warming. The emulator absorbs whatever average relationship held across the seven markers and cannot separate it.

Carbon cycle feedbacks. Permafrost, weakening land and ocean sinks, and anything else that makes the airborne fraction depend on the path rather than the total. Whatever those contributed in the seven FaIR runs is baked into the fitted curve at their average.

Uncertainty. One number is shown, not a range. The FaIR ensemble the fit came from spans well over a degree at any given cumulative total. The emulator reports its central behaviour.

The timing of warming. Only 2100 is reported, not the path to it or anything after it.

Internal consistency. Nothing checks that the six assumptions belong together. The coherence flags name a few combinations no marker contains, worded as an observation rather than a block, because an unexamined combination is not an impossible one. Nothing checks that the energy system implied by a given efficiency and fuel mix could be built, financed or fuelled.

The country comparison

The tile names the **largest** economy whose 2024 CO₂ per dollar of GDP sits within 5% of the reader's 2100 world, where largest means GDP, the denominator of the ratio being compared. It uses Energy Institute CO₂ over World Bank purchasing power GDP, restricted to 66 economies above 25 Mt CO₂ and \$40 billion, so a small or poorly measured economy cannot become the answer. It compares one ratio and nothing else: it does not say the world would resemble that country in any other respect.

Taking the nearest economy of any size, which the tile did until 2026-09-07, kept returning whichever small economy happened to sit on the number: a 2100 world at 0.126 kg/\$ came out as New Zealand, a \$256 billion economy, while the Netherlands sat at 0.124 with five times the output. CMIP7 HIGH was one of the scenarios it hit. The 5% band is what keeps the likeness true; a wider one buys larger names at the cost of the claim, since at 10% a world at 0.16 kg/\$ would "look like" Japan at 0.170.

Where no economy sits within 5%, the tile says so rather than naming one. It reads "no economy this clean", "no close match" or "no economy this carbon-intensive", and the note names the economy at the edge of the table so the reader keeps a bearing. Falling back to the nearest of any size, as it did until 2026-09-07, put Switzerland against a 2100 world at 0.011 kg/\$ when Switzerland emits 0.043 — four times as much. Three of the eight presets land in that stretch, CMIP7 MEDIUM-to-LOW, CMIP7 VERY LOW and the Ausubel methane economy, so the tile was asserting a likeness that did not exist on every one of them.

Four stretches inside the tool's reachable range hold no economy within 5%: below 0.041 kg/\$, and the gaps at 0.046–0.049, 0.071–0.076 and 0.381–0.404. The note prints the reader's own ratio in every case, so the gap stays visible. `tests/analogue.test.ts` walks the whole range and holds every rule.

The chart axis

The vertical axis is not fixed. It is chosen for each render to cover the reader's path and all seven markers together, on round steps, with zero always on it. Two things follow. A scenario that runs far above the published range stays fully drawn rather than clipping at the top of a fixed axis. And the seven markers stay on the same axis as the reader's line, so the comparison holds however far the sliders are pushed, at the cost of the markers compressing when the reader's path dwarfs them.

Rounding

Warming is shown to two decimal places, never more. Cumulative CO₂ is shown as a whole number. Rates carry their sign, so +1.91 and −1.43 read as a pair.

Part 3

Data

Every number the interface shows comes from `src/data/*.json`. Nothing is typed into the UI code. This file says where each series came from, when, and in what units.

Two scripts write those files, split by whether the numbers can be rebuilt from a primary source on this machine.

Script	Writes	Rebuildable
<code>scripts/build_data.py</code>	<code>observed.json</code> , <code>analogues.json</code> , <code>base.json</code>	Yes, from the sources below
<code>scripts/build_wpp.py</code>	<code>learn_population.json</code>	Yes, from the UN download
<code>scripts/build_energy_intensity.py</code>	<code>learn_energy_intensity.json</code>	Yes, from the sources below
<code>scripts/build_fuel_mix.py</code>	<code>learn_fuel_mix.json</code>	Yes, from the sources below
<code>scripts/build_income.py</code>	<code>learn_income.json</code>	Yes, from the sources below
<code>scripts/build_methane.py</code>	<code>learn_methane.json</code>	Yes, from the EDGAR download
<code>scripts/build_land_use.py</code>	<code>learn_land_use.json</code>	Yes, from the cached Global Carbon Budget
<code>scripts/extract_prototype.py</code> then <code>scripts/build_carried_data.py</code>	<code>config.json</code> , <code>emulator.json</code> , <code>markers.json</code> , <code>population.json</code> , <code>presets.json</code> , <code>notes.json</code>	No, carried from the prototype

Sources

Energy Institute Statistical Review of World Energy 2026

Downloaded by hand each June, held at `/home/rpielke/EI-Stats-Review-2026.xlsx` outside this repository because of its size. Sheets used:

- CO₂ from Energy, world row, 1965 to 2024, million tonnes CO₂. Covers CO₂ from energy only, with no cement or other industrial process emissions.
- Total Energy Supply (TES) -EJ, world row and country rows, 1965 to 2024, exajoules.
- CO₂ from Flaring, world row, 1975 to 2024, million tonnes CO₂. Used only in a base-year check.

The 2026 edition carries a 2025 estimate. It is excluded; the series stop at 2024, the last complete year.

One trap. The header row repeats the final year label three times: once for the value, once for the year-on-year percentage change, and once for the share of world total. Reading the header into a dictionary keyed by year silently returns the share column, which is 1.0 for the world row. `read_sheet` cuts the year block at the first non-monotonic repeat and asserts the world value is not 1.0.

World Bank

Fetches live from the public API, no key needed, cached in `scripts/_worldbank_cache.json`.

- `NY.GDP.MKTP.PP.KD`, GDP at purchasing power parity, constant 2021 international dollars. World aggregate 1990 to 2024, and one 2024 value per country for the analogue table. The series begins in 1990, which is why every GDP-derived rate starts there.
- `SP.POP.TOTL`, world population. This is the UN World Population Prospects 2024 revision as redistributed by the World Bank.

Country names in the Energy Institute workbook are joined to World Bank ISO3 codes through `scripts/country_iso3.py`, carried over from the decarbonization project where it was checked against the World Bank country list. It covers 78 of the 80 country rows. Taiwan and USSR have no entry, deliberately: the World Bank publishes no GDP series for either. Two names in that table have caused silent drops before and are handled explicitly, Türkiye (renamed from Turkey in the 2026 edition) and Russian Federation.

Venezuela drops out of the analogue table because the World Bank has no 2024 PPP GDP for it. The build prints every drop rather than swallowing it.

Global Carbon Budget

Fetches from the Our World in Data redistribution (<https://raw.githubusercontent.com/owid/co2-data/master/owid-co2-data.csv>), world rows only, cached in `scripts/_gcb_cache.json`.

Used for the base year, because the CMIP7 markers count all fossil and industrial CO₂ while the Energy Institute series covers energy and flaring alone. For 2024:

Component	GtCO ₂
Fossil and industry, total	38.599
of which cement	1.473
of which flaring	0.416

Component	GtCO2
of which other industry	0.424
Land use change	4.586
Total CO2	43.185

That total is the basis the marker paths are on, which is what the marker values for 2025 confirm: they span 42.4 to 43.8 with a mean of 43.08.

ScenarioMIP CMIP7 marker runs

The seven marker emissions paths, their cumulative totals, their 2100 methane and land use values, and the Kaya factors that place each marker's tick under each slider. Five-yearly from 2025 to 2100, GtCO2 including land use.

No primary source on this machine. These came into the prototype from ScenarioMIP and are lifted out of it by `scripts/extract_prototype.py`. Three markers (LOW, LOW-to-NEGATIVE, VERY LOW) publish no carbon-intensity rate, so those sliders carry four scenario ticks rather than seven.

UN World Population Prospects 2024

Two files, both fetched from the UN Population Division's own download area and both keyed to the 2024 revision.

- `WPP2024_TotalPopulationBySex.csv.gz`, dated 13 December 2024, about 17 MB compressed. Every location, every variant, 1950 to 2100, population in thousands. Used for the world series, the low, medium, high, momentum and constant-fertility variants, the 95% prediction interval, and the seven regional totals the population builder adds up.
- `WPP2024_Demographic_Indicators_Medium.csv.gz`. Total fertility for the world in 1950, 1990, 2024 and 2100, and for each of the seven regions in 2024.

Neither file is committed. `scripts/build_wpp.py` extracts the subset the site uses into `scripts/_wpp_cache.json`, which is committed, so a build works offline. Re-run with `--refresh` to fetch again.

One check the build enforces. The seven regions have to reproduce the UN's own world figures at the low, medium and high variants, to within 0.002 billion. They do: 6.987, 10.180 and 14.395 billion. The build exits rather than write a file whose regions sum to something the UN never published.

The 95% prediction intervals do not add up that way and the site says so on the page: the regional lower bounds sum to 8.234 billion against a world lower bound of 9.047, because the regions do not all land at the bottom of their own ranges at once.

Africa is split into sub-Saharan Africa (an SDG region in the UN's own hierarchy) and northern Africa (a subregion), which sum exactly to the UN's Africa figure at every variant. That keeps the region carrying most of the remaining growth on a control of its own.

IPCC 2006 Guidelines, default carbon content

Volume 2 (Energy), Chapter 1, Table 1.3, read from the published PDF. Values in kilograms of carbon per gigajoule, converted to CO₂ at 44/12:

Fuel	kgC/GJ	kgCO ₂ /GJ
Other bituminous coal	25.8	94.6
Crude oil	20.0	73.3
Natural gas	15.3	56.1

Nuclear, hydro, wind and solar carry none. Bioenergy carries none in energy-sector accounting, where biogenic CO₂ is reported under land use, which is also how the Energy Institute's CO₂ series treats it.

Applied raw, these overstate what the world emits. The 2024 mix gives 65.15 kgCO₂ per GJ against the 59.94 the Energy Institute reports from energy that year, mostly because part of the oil supply becomes plastics, lubricants and bitumen rather than exhaust. `build_fuel_mix.py` records the ratio, 0.9201, and the page applies it so today's mix reproduces today's intensity.

Maddison Project Database 2023

Downloaded from the Dataverse copy the Groningen Growth and Development Centre publishes, cached as `scripts/_mpd2023.xlsx` (5 MB, not committed) with the extracted world series in `scripts/_maddison_cache.json` (committed).

Used for one thing: world output before 1990, where the World Bank's purchasing-power series does not reach. The `Full` data sheet gives GDP per capita and population by country and year. The build sums GDP over the **151 countries with an unbroken 1965 to 2022 record**, 96.5% of world GDP in 1990, so each year compares the same economies as the last rather than a growing sample.

Only the growth rates are used. The World Bank level in 1990 is carried backwards on them, so every level from 1990 onward rests on the World Bank alone.

How much that choice decides. Over the 32 years the two sources share, the Maddison sample grows 3.46% a year and the World Bank series 3.19%. Carrying the level back on growth rates adjusted to close that gap gives a 1965 energy intensity of 5.86 MJ per dollar instead of 6.25, and a whole-record improvement of −0.92% a year instead of −1.03%. Both figures appear on the page.

IIASA SSP database v3.2 (June 2025 release)

World population trajectories for SSP1, SSP2 and SSP3, five-yearly 2025 to 2100, billions, aggregated from countries to world. Also carried from the prototype with no primary source here.

The UN 2024 95% prediction interval for 2100 population, 9.0 to 11.4 billion around a median of 10.2, is stated in the brief.

FaIR v2.2

The emulator coefficients, fitted to FaIR runs of the seven markers using the fair-calibrate v1.4.1 constrained ensemble. Carried from the prototype. See Part 2 for what the fit does and does not reproduce.

The files

base.json

The base-year state, all from 2024, the last complete year, applied at 2025.

Field	Value	Units	Source
populationBn	8.1409	billions	World Bank SP.POP.TOTL
gdpPerPersonUsd	21,393.2	constant 2021 international \$	World Bank PPP GDP over population
energyPerDollarMj	3.4002	MJ per dollar	EI total energy supply over World Bank PPP GDP
co2PerEnergyKgGj	65.181	kg CO2 per GJ	Global Carbon Budget fossil and industry CO2 over EI total energy supply
landUseGt	3.83	GtCO2/yr	stated in the brief
methaneMt	380	Mt/yr	stated in the brief

superseded records what the prototype used and why it changed. The old value of 60.5 kg per GJ came from energy CO2 plus flaring alone, which started every path about 4.4 GtCO2 below all seven markers.

Land use stays at the brief's 3.83 rather than the Global Carbon Budget's 4.59 for the same year. The gap sits well inside the uncertainty on that term, unlike the cement omission, which was definite.

observed.json

World history 1965 to 2024, the three Kaya factors derived from it, and the rate windows the notes quote.

Rate	Value	Period
Income per person	+1.90 %/yr	1990 to 2024
Energy per dollar	−1.43 %/yr	1990 to 2024
CO2 per unit energy	−0.21 %/yr	1990 to 2024
Income per person	+2.02 %/yr	2015 to 2024
Energy per dollar	−1.62 %/yr	2015 to 2024
CO2 per unit energy	−0.48 %/yr	2015 to 2024

Window extremes, each reported with the window that produced it:

Window	Highest	Lowest
Income, 25 years since 1990	+2.23 %/yr (1994–2019)	+1.86 %/yr (1990–2015)
Energy per dollar, 25 years since 1990	−1.36 %/yr (1990–2015)	−1.57 %/yr (1996–2021)
CO2 per unit energy, 30 years since 1965	−0.17 %/yr (1992–2022)	−0.56 %/yr (1965–1995)

Reported as plain highest and lowest rather than slowest and fastest, because which end counts as slow progress flips with the sign of the series.

learn_population.json

Everything the population Learn More page draws and builds from, in the shape every Learn More data file uses: series for lines, bands for uncertainty, parts for builder controls, constants for the numbers the prose quotes.

Field	Value	Units
World, 2025	8.232	billions
World peak, medium	10.289 in 2084	billions
World 2100, medium	10.180	billions
World 2100, 95% interval	9.047 to 11.437	billions
World 2100, low and high variants	6.987 and 14.395	billions
World 2100, momentum variant	9.596	billions
World 2100, constant fertility	18.193	billions
Total fertility, world	4.85 (1950), 3.31 (1990), 2.25 (2024), 1.84 (2100)	births per woman

The seven builder regions, 2100, billions:

Region	Low	Medium	High	2025
Sub-Saharan Africa	2.378	3.351	4.606	1.274
Northern Africa	0.319	0.462	0.648	0.276
Asia	3.074	4.613	6.677	4.835
Europe	0.413	0.592	0.824	0.744
Latin America and the Caribbean	0.400	0.613	0.905	0.668
Northern America	0.348	0.475	0.637	0.388
Oceania	0.053	0.073	0.098	0.047

Not yet here: IHME. The brief asks for the IHME projection alongside the UN and the SSPs on that chart. The IHME data portal was unavailable when the page was built, so the chart carries a line saying the projection joins it in a later revision. Nothing on the page reports an IHME number.

learn_energy_intensity.json

World energy intensity 1965 to 2024, and the rate of every 25-year window inside it.

Field	Value	Units
Intensity, 1965	6.2506	MJ per dollar
Intensity, 1990	5.5442	MJ per dollar
Intensity, 2024	3.4002	MJ per dollar
Rate, 1965 to 2024	−1.03	%/yr
Rate, 1990 to 2024	−1.43	%/yr
Rate, 2015 to 2024	−1.62	%/yr
Primary energy, 1965 and 2024	149.7 and 592.2	EJ
Energy growth, 1965 to 2024	+2.36	%/yr
Output growth, 1965 to 2024	+3.42	%/yr

The 2024 intensity of 3.4002 MJ per dollar reproduces base.json exactly, and the 1990 to 2024 rate of −1.4277 reproduces observed.json, because both rest on the same World Bank series over those years.

The 35 windows of 25 years run from **−1.5675 %/yr (1996 to 2021)** to **−0.4786 %/yr (1965 to 1990)**, with a median of −1.07. CMIP7 HIGH's −0.66 sits slower than 31 of the 35. The tool's own "Trend continues" rate of −1.62 sits faster than all 35.

learn_fuel_mix.json

Shares of world primary energy by fuel, 1965 to 2024, the emission factors above, and the two carbon-intensity series.

Share of primary energy	1965	2024
Coal	38.9%	27.9%
Oil	43.3%	33.6%
Gas	15.2%	25.1%
Nuclear	0.2%	5.2%
Hydro	2.2%	2.7%
Wind and solar	0.0%	2.9%
Bioenergy and other	0.2%	2.6%

Whatever the named fuels miss, biofuels most of it, joins bioenergy and other, so the shares add to the total energy supply exactly. The build exits if they do not sum to 100 in any year.

Two intensities, and they differ. On the basis the slider measures, which includes cement, flaring and other industrial CO2 as the CMIP7 markers do, the world ran 75.56 kgCO2 per GJ in 1965 and 65.18 in 2024. On combustion alone, 74.79 and 59.94. The 2024 slider-basis figure reproduces base.json exactly.

That gap changes the rate as well as the level: **−0.15%/yr from 1990 to 2024 on the slider's basis, −0.21%/yr on combustion alone.** The calibration mark under the front page's slider used −0.21 until 2026-09-05 and now uses −0.15, so the mark and the slider measure the same quantity; config.json keeps the old figure under supersededRates. Part 2 records what moved with it.

The non-combustion term, 5.237 kgCO2 per GJ in 2024, breaks down as cement 2.487, other industry 0.717, flaring 0.702, and 1.331 for the difference between the Energy Institute and Global Carbon Budget inventories. Holding it where it stands puts a floor of 5.237 kgCO2 per GJ under any fuel mix, which caps the improvement this page can reach at −3.31%/yr.

learn_income.json

World GDP per person 1965 to 2024, on the same spliced output series as the energy-intensity page, divided by World Bank world population, plus the three World Bank income groups from 1990.

	Level, 2024	Growth since 1990	Past decade	Share of people	Share of output
World	\$21,393	+1.90 %/yr	+2.02 %/yr		

	Level, 2024	Growth since 1990	Past decade	Share of people	Share of output
High income	\$58,645	+1.53 %/yr	+1.53 %/yr	17.4%	47.8%
Middle income	\$15,025	+3.51 %/yr	+3.21 %/yr	73.4%	51.5%
Low income	\$2,396	+0.75 %/yr	+0.22 %/yr	9.2%	1.0%

The 2024 world figure reproduces base.json exactly. The three groups cover 100.0% of world population and their population-weighted average comes to \$21,464, 0.3% above the world figure; the builder scales by that ratio.

The current World Bank classification is applied across the whole record, so no country moves group mid-series.

learn_methane.json

World anthropogenic methane by source, 1990 to 2023, from EDGAR's 2024 release, summed from country and IPCC 2006 sector to five groups.

Source	1990	2023	Share	Growth
Fossil fuels	87.6 Mt	118.7 Mt	34.0%	+0.93 %/yr
Livestock	102.2 Mt	125.5 Mt	35.9%	+0.63 %/yr
Rice	39.1 Mt	36.8 Mt	10.5%	-0.19 %/yr
Waste	42.7 Mt	65.5 Mt	18.7%	+1.31 %/yr
Other anthropogenic	1.7 Mt	3.1 Mt	0.9%	+1.87 %/yr
Total	273.2 Mt	349.6 Mt		+0.75 %/yr

Rice is the one source that fell.

EDGAR's inventory and the tool's base year differ. EDGAR reaches 350 Mt for 2023; the tool uses 380 Mt. The Global Methane Budget 2000–2020 puts direct anthropogenic emissions at 369 Tg a year for 2010–2019, range 350 to 391, so both numbers sit inside the published range. The page scales each source by 1.086 so today's five reproduce the tool's base.

Natural wetlands, which the Global Methane Budget puts at 248 Tg a year together with inland fresh water, sit outside the slider and outside this file.

learn_land_use.json

The Global Carbon Budget land-use flux 1965 to 2024 with its uncertainty, its decomposition, and the forest growth rates the restoration control uses.

Field	Value
Flux, 1965	6.39 GtCO2/yr
Flux, peak	8.02 GtCO2/yr in 1997
Flux, 2024	4.59 GtCO2/yr
Uncertainty	±2.57 GtCO2/yr (±0.7 GtC, 1σ)
Gross deforestation, 2014–2023	6.23 GtCO2/yr
Regrowth, 2014–2023	4.40 GtCO2/yr
Other transitions and peat	2.20 GtCO2/yr (the remainder)
Net, 2014–2023	4.03 GtCO2/yr

Two figures for the same decade. The Our World in Data redistribution drawn on the chart averages 5.21 GtCO2 over 2014–2023; the Global Carbon Budget 2024 paper reports 4.03 for that decade. The 1.18 GtCO2 between them, most likely a vintage difference, sits well inside the ±2.57 uncertainty. The page says so rather than choosing one silently.

Forest growth rates come from IPCC 2006 Volume 4, Table 4.9, in tonnes of dry matter a hectare a year, converted at the 0.47 carbon fraction (Table 4.3) and 44/12:

Stand	t d.m./ha/yr	tCO2/ha/yr
Tropical rain forest, South America, over 20 years	3.1	5.3
Young regrowth, Africa	10.0	17.2
Young regrowth, South America	11.0	19.0
Young regrowth, insular Asia	13.0	22.4

Above-ground biomass only; below-ground carbon and soil add more.

analogues.json

66 economies, their 2024 CO2 and PPP GDP, and CO2 per dollar in kilograms. Restricted to economies above 25 Mt CO2 and \$40 billion. The build asserts the count and prints anything dropped at the join. GDP carries the comparison as well as the ratio: the tile names the largest economy within 5% of the reader's 2100 world, not the nearest of any size, and says plainly that there is none when nothing sits that close. See Part 2.

markers.json, population.json, emulator.json, config.json, presets.json, notes.json
Carried over, as above. config.json also keeps prototypeBase, the base-year state the prototype used, so the two can be diffed.

Vintages

Source	Vintage	Checked
Energy Institute Statistical Review	2026 edition	2026-09-04
World Bank PPP GDP and population	live API	2026-09-04
Global Carbon Budget via Our World in Data	live file	2026-09-04
ScenarioMIP CMIP7 markers	as carried in the prototype	2026-09-04
IIASA SSP database	v3.2, June 2025 release	2026-09-04
UN World Population Prospects	2024 revision, file dated 2024-12-13	2026-09-05
Maddison Project Database	2023 release	2026-09-05
EDGAR greenhouse gases	2024 release, CH4 to 2023	2026-09-05
Global Methane Budget	2000–2020, published 2025	2026-09-05
Global Carbon Budget paper	2024, published 2025	2026-09-05
IPCC 2006 Guidelines	Volumes 2 and 4	2026-09-05
FaIR calibration	v2.2, fair-calibrate v1.4.1	2026-09-04